The empirical recurrence for OEIS A235977: recovering a conjecture that is not written down

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Abstract

OEIS A235977 records "Empirical recurrence of order 61", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 69$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 61, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 61 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235977 is "Number of (n+1)X(4+1) 0..2 arrays with the upper median equal to the lower median in every 2X2 subblock.". It has offset 1 and begins

$$3741, 89555, 1947197, 46727365, 1101519021, 26513929449, 636000634655, 15322368309233, \ldots$$

The entry states:

> Empirical recurrence of order 61 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 08:58:59, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 69$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 69$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 138$ exact terms of the model returns order 61 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{61} c_i \, a(n-i),$$

c_{1}	44	c_{17}	−21274199278095097	c_{33}	51126479078376159142790	c_{49}	−1147201590
c_{2}	−224	c_{18}	46273693558558339	c_{34}	−955417376458134206024	c_{50}	−220602793
c_{3}	−13789	c_{19}	269480971841331925	c_{35}	−95944975709864960530940	c_{51}	1752351708
c_{4}	176699	c_{20}	−940331199376719096	c_{36}	42009239314920659888722	c_{52}	782397797
c_{5}	1343212	c_{21}	−2179222322993189523	c_{37}	119122083899496717851840	c_{53}	−158740351
c_{6}	−32156344	c_{22}	12017652087154925417	c_{38}	−94941715224992269281545	c_{54}	53492434
c_{7}	1989435	c_{23}	8678187285829499584	c_{39}	−92691864926811514259610	c_{55}	82970919
c_{8}	2826502255	c_{24}	−104834938822675165743	c_{40}	114791321966881830143686	c_{56}	−6147760
c_{9}	−9975773577	c_{25}	25117509735614808691	c_{41}	37947109080271802741518	c_{57}	−2318089
c_{10}	−138104246395	c_{26}	637132107998800338013	c_{42}	−85636495357712422030732	c_{58}	2004076
c_{11}	874153923559	c_{27}	−605480282254364848559	c_{43}	−123052968013214523944	c_{59}	2965665
c_{12}	3721547981341	c_{28}	−2667342592327312653546	c_{44}	40585250343844763897616	c_{60}	−202755
c_{13}	−39658646727305	c_{29}	4310482543912176147185	c_{45}	−8252558195870627605736	c_{61}	−11680
c_{14}	−37153908483318	c_{30}	7301144475279762186839	c_{46}	−12133804998261259984048		
c_{15}	1126565430750085	c_{31}	−18292307092222131724400	c_{47}	4355432201980957262592		
c_{16}	−973592362266889	c_{32}	−10885387961800567505266	c_{48}	2205914641025845147936		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 61, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $61 + 4$ terms removed returns order 61 as well, so $\deg q_{\min} = 61 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $61 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 61 that this sequence satisfies — and that family turns out to have exactly one member.

References

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